Proxy Embeddings: Steering and Measuring Diversity Through a Space That Is Not the Artifact's

Abstract

Every diversity objective in this line of work is computed in an embedding, and the embedding is a proxy: for a rendered image it is a text encoder's view of the instruction, for a poem it is a semantic model nearly blind to metre, for an exam item it is a space in which the answer key does not exist. Recursive Axis Conditioning (RAC) assumes an embedding oracle and no inverse; this paper asks what follows when the oracle is a proxy for the artifact rather than the artifact itself, and extends the method across that gap.

We measure the gap in three modalities. Pairwise similarity of image instructions in text-embedding space predicts pairwise similarity of the rendered pictures at **Pearson *r* between 0.167 and 0.421** across seven arms — 14.5% shared variance pooled, 8.2% within-arm — and instructions with a 0.000 duplicate rate render to one visual mode. In audio, **whether a prompt can be steered before rendering is a property of the embedder**: prompt-to-track alignment is 0.68 under MuQ-MuLan and 0.18 under CLAP-music, per-arm verdicts flip between embedders, and each nominates a different most-redundant pair. Conditioning itself degrades at the seam where one model's language becomes another model's input: **nine of eleven image axes are realized in the written instruction and four of eleven in the render**, a mean loss of 35%. And an embedding is blind to every property it does not encode, including ones that decide usability — **90% of a generated exam bank's keys sit in position A** and no diversity measure moves.

Three things survive the gap. *Steering through the proxy*: rendering a bounded sample, embedding it in the artifact's own space and mapping its crowded directions back into the text space gives the best arm in the image domain (+18% centered Vendi over text-only steering at matched *n*) and raises mean-centered CLAP Vendi on rendered instrumentals from 11.5 to 17.43. *Selecting on the channel the objective cannot see*: best-of-3 renders chosen on pixel statistics raise optical min-gap 1.24× at no cost in CLIP. And *the ranking does not depend on the viewpoint*: under four independent representations, three of which the method never optimizes, RAC ranks first under every one, at a mean Kendall τ of +0.83. The prescriptions are short: name the embedder, measure at the artifact, audit in a channel the objective never optimizes, and balance by construction what the embedding cannot see.

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1. Introduction

Recursive Axis Conditioning (RAC) is a generation loop that improves a chosen diversity measure: it asks the generator to name the language-valued axes along which its own outputs can differ, ranks them, takes their most-different levels, splits an exhausted axis into finer sub-axes, and keeps one of K candidates by the measure. The companion paper, *Recursive Axis Conditioning for Diverse Synthetic Data Generation*, builds that loop, measures it against the released state of the art — first of twelve corpora on coverage of a held-out human reference at a twentieth of Alpaca's budget, an exam bank with no enemy pair at a blind-adjudicated radius — and states the theory that bounds it. Everything in it rests on one assumption: an *embedding oracle E* that maps an item to a point in $\mathbb{R}^D$, with no inverse. Every axis is scored in *E*, every candidate is selected in *E*, and every diversity number is a statistic of *E*.

This paper is about the word *oracle*. In two of the companion's five domains the thing being made is not text at all. An image instruction is written by one model and rendered by another; a music prompt becomes a two-minute instrumental. The text embedder never sees the picture or hears the track, so what it measures is a *proxy* for the artifact the reader receives — and even where the artifact is text, `nomic-embed-text` is a semantic model nearly blind to metre, rhyme and line length, and no embedding at all encodes where an exam item's answer key sits among its options. The oracle is always a proxy. The question is how far it is from the artifact, what falls into the gap, and what a method that only ever sees the proxy can do about it.

We answer in four parts. **What the proxy sees** (§4): text-embedding similarity between image instructions predicts rendered-image similarity at Pearson *r* between 0.167 and 0.421, so most of the visual variance is unexplained by anything a text-side method optimizes; in audio, whether a prompt can be steered before rendering is a property of the embedder, with prompt-to-track alignment of 0.68 under MuQ-MuLan against 0.18 under CLAP-music; literal and latent diversity of one corpus move in opposite directions as it grows; a whole class of visual repetition — tiled grids, shared palettes, recoloured compositions — is invisible to embedding metrics and visible to a 16×16 luminance map; and a generated exam bank whose keys sit 90% in position A scores identically on every diversity measure to a balanced one. Against that, the *ranking* of methods does survive the choice of viewpoint: under four independent representations, three of which the method never optimizes, RAC ranks first under every one. **Steering through the proxy** (§5): rendering a bounded sample, embedding it in the artifact's own space and mapping its crowded directions back into the text space gives the best image arm at matched *n* and, through an audio embedder, raises mean-centered CLAP Vendi on rendered tracks from 11.5 to 17.43; and when a pipeline pays for extra renders, choosing among them on the channel the objective cannot see buys 1.24× on that channel at zero cost in CLIP. **Conditioning across the seam** (§6): by intervention, nine of eleven image axes are realized in the written instruction and four of eleven in the render, a mean loss of 35% of the effect at the boundary where one model's language becomes another model's input. **Scoring in the artifact's space** (§7): the obvious way to make axis scoring empirical — replace level descriptions with the centroid of what each level produced — is a marginal mean where a partial effect is needed, and it is the companion's manipulation-based repair that makes artifact-space scoring usable at all.

The contributions, in the order we would defend them:

- **A measured proxy gap in three modalities** (§4.2, §4.3), with the pooled and within-arm shared variance stated separately and the embedder named in every audio number.
- **Cross-modal steering as an extension of the loop** (§5.1, §5.2): the same ledger that repels what the model keeps *saying* also repels what it keeps *showing*, through a least-squares bridge from the artifact's embedding into the text embedding the loop already optimizes.
- **The seam result** (§6): conditioning strength degrades where the generator changes hands, established by manipulation rather than correlation, with the instrument's share of the loss separated from the renderer's.
- **Two failures no embedding can see** (§4.6, §4.7), with the audit that finds each and the repair that needs no embedding — a structural ban block for compositional repetition, and a uniform key permutation that needs no answer key.
- **Robustness of the ranking to the viewpoint** (§4.4): four independent representations agree at a mean Kendall τ of +0.83, and the method the companion paper proposes ranks first under all four.
- **A short practice** (§8): name the embedder, measure at the artifact, audit in a channel the objective never optimizes, and balance by construction what the embedding cannot see.

2. The oracle is a proxy

We assume throughout:

> **Assumption (embedding oracle, no inverse).** We have access to E: Text $\to \mathbb{R}^D$, computable on demand. We have **no** E^{-1}: $\mathbb{R}^D \to$ Text.

This is the structural fact that shapes the entire design. Given a corpus X_n we can compute, exactly and cheaply, the point in $\mathbb{R}^D$ that would most improve any of our diversity measures — the direction of least occupied spectral energy, the centre of the largest empty ball, the point maximizing marginal coverage. Knowing that point is worth nothing on its own, because no procedure turns a target embedding back into a poem.

Consequently the system cannot *solve* for its next item. It can only **propose** (sample from $p(\cdot \mid x)$ for some prompt x we can write), **measure** (embed the proposals and score them), and **select** (keep one). All of the control is in the choice of x, and that choice is expressible only in language. This is why the latent variables here are language-valued — named axes with named levels — rather than continuous codes: they are the only handles that reach the generator. It is also why the axis scoring of the companion paper's §5 exists. **The axis scoring is a surrogate for the missing inverse**: unable to decode the direction we want to travel, we score the language-valued conditions we *can* write by how nearly their induced output distributions point that way.

Three consequences follow for measurement, and they order the evidence in this paper.

Two facts about circularity are worth stating before the results. Our packing selection rule maximizes a weighted sum of embedding-space orthogonality and embedding-space min-gap, and we then report embedding-space diversity metrics; centered Vendi is a monotone function of how flat the Gram spectrum is, which is close to what the orthogonality term climbs, and median nearest-neighbour distance *is* the min-gap term. Those wins should be read as confirmation that the optimizer works, not as independent evidence. The independent evidence is the literal measures, which the method never observes, and the out-of-objective channels of §4.1, which live downstream of a rendering step or in a feature space nothing in the loop touches. Sorted by how much weight they can bear: embedding-space measures are partly circular; literal-space measures are independent; rendered-artifact and out-of-objective measures are the most independent, and they carry the argument.

The proxy adds a fourth rung below all three. When the artifact is rendered by a model the embedder never sees, even the most independent text-side measure is a measure of the *instruction*, and whether the instruction's diversity reaches the picture is an empirical question — with, it turns out, a discouraging answer (§4.2). The companion paper states five practices that follow from naming the objective; this paper is where the last three are tested.

1. **Say which objective you mean.** They are different problems with different optimal policies, and ”diverse” does not distinguish them.
2. **Count exact duplicates before computing anything.** A 74% duplicate rate makes every distance statistic a statistic about duplication.
3. **Publish the pairwise-similarity distribution** your kernel operates on. Mean pairwise cosine was 0.883 in one of our corpora and 0.444 in another under the same embedder; an ε or a Vendi score means nothing without it.
4. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven rendered arms, and 8.2% within-arm (the companion paper's §4.2).
5. **Audit in a channel the objective never optimizes** (§4.1, §4.7; the companion paper's §8.3). It is the only measurement in this paper that cannot be a restatement of the selection rule, and it is where both the strongest positive result and the largest undetected defect were found.

3. Setup

Generator and embedders. `openai/gpt-5.6-luna` via OpenRouter for generation, judging, attractor mining and axis elicitation. Text embeddings: `nomic-embed-text` (768-d, unit-normalized) served locally by Ollama, so embedding is free and the loop is never rate-limited by its own measuring instrument. Images: `gpt-image-1-mini`, embedded with CLIP ViT-B/32. Audio: Lyria 3 Pro instrumentals, embedded with CLAP and MERT. Every corpus is append-only JSONL with embeddings checkpointed alongside, resumable after a kill; every number carries a provenance tag naming the procedure that produced it.

Domains. *DALL·E instructions for post-modern artworks*, where diversity is the product and the same corpus can be measured in three spaces — words, text-embedding, and the rendered picture. *Psychometric test questions*, multiple-choice items assessing general reasoning, where two items with the same construct are a validity problem rather than an aesthetic one. *Instruction corpora*, for the head-to-head against released artifacts. *Poetry*, a pilot where craft rather than content carries the variation. *Instrumental-music prompts*, the longest chain in the paper: latent axes → text prompt → two-minute instrumental → audio embedding.

Methods compared. Five published approaches, implemented faithfully rather than as strawmen, each getting the same generator, embedder and budget accounting as ours.

arm	what it is
`naive`	plain repeated prompting — the honest floor
`high_temp`	temperature 1.6 — the trivial diversity lever
`self_instruct`	**Self-Instruct / Alpaca**: few-shot exemplars sampled from the pool, plus ROUGE-L ≤ 0.7 rejection against it
`evol_instruct`	**Evol-Instruct (WizardLM)**: sample an existing item, apply a random evolution operator (deepen, concretize, add constraint, harder reasoning, mutate form)
`persona`	**Persona-Hub / AttrPrompt**: a flat catalogue of personas × attributes, sampled uniformly
`rac`	**Recursive Axis Conditioning** (ours)
`rac+vision`	RAC, plus steering on the *rendered image* (the companion paper's §4.2)

What we measure, and what each measure misses. We report several because they disagree, and a corpus that one of them calls healthy another calls collapsed.

- **Exact-duplicate rate** — byte-identical repetition. Unambiguous, cheap, and the first thing to compute; blind to paraphrase.
- **Distinct-*n*, 4-gram self-repetition, *n*-gram Vendi** — surface statistics over token sequences. They catch templating that the duplicate rate misses and read meaning not at all. **These are independent of our objective**, since the method sees only embeddings and has no access to token counts or string identity.
- **Mean-centered embedding Vendi** — the exponential of the von Neumann entropy of the corpus Gram matrix, an effective number of distinct items. It summarizes the whole spectrum, which makes it insensitive to local structure. Centering matters as much as the statistic: the shared mean direction of text embeddings compresses the uncentered score by roughly 2.7×, so an uncentered Vendi reports the cone as much as the content.
- **Median nearest-neighbour distance** — how much room the typical item has. It goes to exactly zero when the median item has a perfect twin.
- **Worst-case nearest-neighbour distance (the packing radius)** — the only measure here under which a single collision is a defect regardless of everything else.
- **Coverage, density, precision and recall against a reference** — computed with reference-side k-NN radii so nothing is tunable per corpus. These are the only measures that know what the space is supposed to look like, and the only ones that let corpora of different scales be compared. Covered fraction at a *fixed* radius does not survive that comparison: it correlates −0.991 with within-corpus spacing, so it ranks corpora by how tightly they cluster rather than by how much they reach.
- **A judged threshold**, where an application defines the failure (the companion paper's §7.4).
- **Measures on the rendered artifact**, and **measures in a channel the loop never optimizes** (§4.1), which are the ones that cannot be circular. **Scale and cost.** The text corpora comprise

43,171 real generations for roughly \$8.50 of OpenRouter spend, plus 1,796 rendered images across sixteen image arms and up to three quality tiers, a further 285 renders for the two render-side probes of §5.3 and §6, 100 rendered Lyria instrumentals, and 236 adjudicated exam-item pairs. Both `naive` arms reach n = 10,000; the reimplemented baselines reach 2,500 each. Every comparison is reported at a matched n that all compared arms actually reached. Total API spend across the project is roughly \$40.

4. What the proxy sees

4.1 Audit in a channel the proxy never sees

The strongest evidence available for any embedding-optimized method is a channel that is *causally downstream of the artifact and upstream of nothing the loop reads*. Each artifact domain has one, and the companion paper's central comparison (its §7.5) is made there: twenty-two prosodic and structural features for poems, where `nomic-embed-text` is nearly blind to metre and rhyme; thirteen item-architecture features for exam items; nine pixel statistics — luminance, contrast, tonal range, saturation, colourfulness, hue entropy, edge density, edge anisotropy, and where the visual weight sits — for renders. At matched generator budget RAC separates from the strongest comparison on all three: 2.25× the prosodic spread of naive prompting on 200 of 200 random draws, 97.6% distinct item architectures against 43.9% after deduplicating both banks, and 1.53× the optical spread with every one of eleven RAC image corpora above every one of five published baselines (U = 55 of 55, p = 0.0002).

What that comparison teaches about instruments is the part that belongs here. **Use a distance statistic, not a spectral one, on a designed feature channel.** Vendi is the exponentiated entropy of a covariance spectrum and scores the effective *dimensionality* of variation rather than its extent; on a 768-dimensional embedding no single direction dominates and the distinction rarely bites, but on a nine-dimensional feature vector it dominates completely, and a corpus that spreads far along two or three optical directions scores *below* one that wobbles slightly along all nine. On mean pairwise distance, optical spread correlates with CLIP diversity at r = +0.76 (p = 0.003) across thirteen corpora; under the spectral estimator the same corpora correlate at −0.60. The proxy and the artifact channel agree or disagree depending on which statistic is read off the channel, before any question about the embedding arises. Degradation does not explain the separation either: blurring k of sixty baseline renders lowers CLIP Vendi monotonically, 43.81 to 41.72 as k runs 0 to 30, so an embedding-space objective is penalized rather than rewarded for admitting degraded images.

The rest of this section is what such channels reveal that the proxy does not.

4.2 Text diversity is nearly blind to image diversity

We rendered the first 199 DALL·E instructions from the naive corpus and embedded them with CLIP. Pairwise cosine similarity in text-embedding space correlates with pairwise cosine similarity in image-embedding space at **Pearson *r* between 0.167 and 0.421** across the seven rendered arms, and at 0.285 within-arm centred. The gap varies by a factor of 2.5 depending on which corpus is measured: lowest on `self_instruct` (0.167) and on the naive arm (0.170), highest on the arm this paper's own method produced (0.421). Pooled, the shared variance is 14.5%; within-arm it is 8.2%. Knowing that two instructions are semantically far apart therefore tells you rather little about whether the two pictures look different, and every text-side diversity method in this literature — ours included — is optimizing a proxy that leaves most of what the reader receives unexplained. That the correlation is strongest on our own corpus is worth stating plainly: the proxy is least misleading exactly where the instructions were built to differ structurally rather than only lexically.

The qualitative version is more damning than the correlation. Independently generated instructions, from a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity, render to near-interchangeable pictures: the same magenta-and-cyan palette, a classical marble bust, neon signage, halftone collage, a receding grid. A vision judge shown samples rates the set's distinctness 6–7 out of 10 and names the attractors precisely — *"muted beige, cream, brown, ochre and black foundations accented by saturated cyan/teal, turquoise, pink"*, *"appropriation of canonical or religious imagery, especially Mona Lisa-like female portraits"*, *"frontal, museum-like presentation with centered, sym-*

metrical compositions". This is a second mode collapse, downstream of ours, contributed by the image model and by the fact that much of what varies in the text lands in the same visual place.

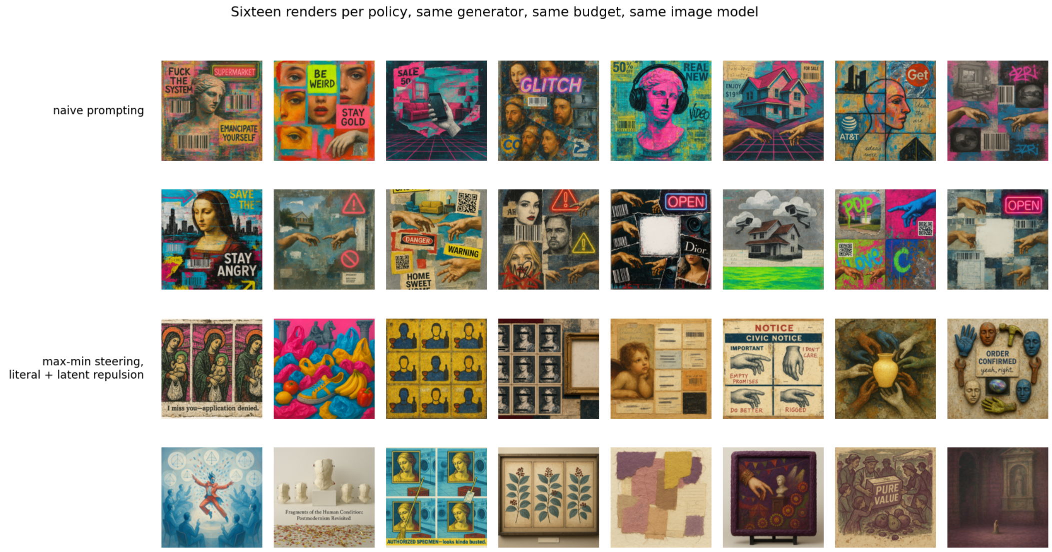

Figure 1. Sixteen renders per policy, same generator, same budget, same image model. Rows 1–2, naive prompting: a corpus with a 0.000 exact-duplicate rate and healthy lexical diversity that still returns one visual mode — magenta and cyan collage, barcodes and QR codes, Renaissance portraits and classical busts, Michelangelo hands, warning triangles and neon OPEN signs recurring across nearly every panel. Rows 3–4, max-min steering with literal and latent repulsion on both the text and vision sides: medium, palette, register and composition all move, across a medieval triptych, a botanical cabinet, a photographed sculpture installation, a torn-paper abstract and a civic notice.

4.3 Audio: whether a prompt can be steered is a property of the embedder

The more portable finding is that whether a prompt can be steered before rendering is a property of the *embedder* rather than of music. Measuring the Spearman correlation between prompt-pair similarity through the text tower and track-pair similarity through the audio tower:

embedder	alignment (naive / conditioned)	note
MuQ-MuLan	**0.68 / 0.47**	a usable steering gradient
CLAP-fused	0.50 / 0.21	weak; ~0.05 on the steered corpus
CLAP-music	0.18 / 0.06	worst; median within-corpus NN distance 0.004 — it hears one track

Cross-embedder agreement on pairwise track similarity spans 0.22–0.84, per-arm diversity verdicts flip between embedders, and each embedder nominates a different most-redundant pair. Two prescriptions follow: steer music with MuQ-MuLan rather than CLAP, and never publish an audio-diversity number without naming its embedder. Structural contracts have a length ceiling as well — compliance with a requested section sequence is exact at prompt lengths near 560 characters, 0.11 at a mean of 658, and zero by ~1,600 — so a structural contract for music must be short or it is noise.

4.4 The ranking does not depend on the viewpoint

A max-min score is a distance in a chosen embedding, so a natural worry is that it names a property of the embedder rather than of the corpus. Six exam-item corpora — this method's bank against naive sampling, persona prompting, self-instruct, evol-instruct and high-temperature sampling, 200 items each — were scored under four independent representations: CLIP text, TF-IDF with no learned semantics at all, the twenty-two-feature prosodic vector, and a 150-dimensional function-word profile of the kind used for authorship attribution. Each score is the fifth-percentile nearest-neighbour distance divided by the mean pairwise distance of the same cloud, which is invariant to the global rescaling an embedding choice is free to apply, and each representation is admitted only if its between-corpus spread exceeds its own within-corpus sampling noise — all four clear that bar by factors of 9.7 to 24.7.

The four views agree, at a mean Kendall tau of +0.83 across their six pairings, and **RAC ranks first under every one of them**, including the three it never optimizes. It also holds the best worst-case score, so it wins under a distributionally-robust reading of diversity as well as under any single view. The agreement is not an artifact of the degenerate baselines: restricted to the three corpora whose fifth-percentile distance is non-zero in every representation, mean tau is +0.67 and RAC is still first under all four. Naive sampling and high-temperature sampling score exactly zero on prosody and on stylometry — at least one item in twenty has an exact neighbour in those channels.

4.5 Literal and latent diversity move in opposite directions

Literal and latent diversity move in opposite directions in one corpus, which is why both are reported throughout. Measuring the DALL·E naive corpus as it grows:

n	distinct-2	4-gram self-repetition	n-gram Vendi	centered embedding Vendi
5	0.870	0.031	4.9	3.79
40	0.492	0.211	30.2	24.13
300	0.252	0.384	142.5	55.70
1,000	0.145	0.520	246.6	63.05
1,750	0.112	0.577	283.3	65.30

Literal diversity collapses monotonically — by n = 1,750 more than half of each new instruction's 4-grams have already appeared — while latent diversity rises and then saturates. Reporting either alone supports an opposite conclusion about the same corpus.

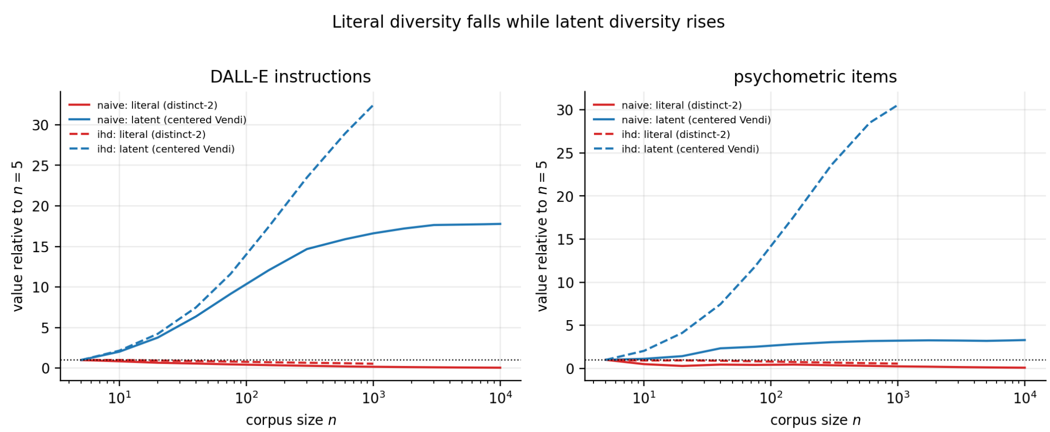

Figure 2. The decoupling, normalized to each series' value at n = 5. Literal diversity falls monotonically while latent diversity rises: two measurements of two different quantities that one word, "diversity", has been covering for.

4.6 A class of repetition the embedding misses

One comparison in the artifact channel runs against the method, and it belongs here. Embedding metrics miss an entire class of visual repetition — tiled grids of one cell, a shared palette, the same composition recoloured — so we audit it with deliberately dumb non-semantic signatures: a 16×16 luminance layout map, an autocorrelation tiling score, a hue histogram. Scored at matched n = 60 and matched render tier, **the published baselines are better than our arms on both structural measures**: layout twins above 0.5 cosine average 0.223 across the five baselines against 0.473 across our eleven, and literal tilings 0.117 against 0.406, with no baseline worse than our best arm on layout. A corpus told to differ along seven or eleven named dimensions apparently converges on a compositional template that plain repeated prompting does not. Adding literal-space structural bans to the prompt — grids banned when recent renders tile, dominant hue pairs named and banned, layout-change demands when layouts collide, plus a learned text→bad-structure bridge — halves palette twins (0.78 → 0.35 in the coverage arm) and improves the coverage objective 21% (0.206 → 0.247), but does not close it. Optical spread and structural repetition are both measured on pixels and point in opposite directions: our corpora occupy a wider region of that space while repeating their compositional scaffolding more often within it.

4.7 A defect no diversity measure can see

The audits of §6 ask whether the axes reach the artifact. This one asks whether the resulting bank is *usable*. For a test bank, key position is a hard requirement: correct answers must be distributed across option positions, because an examinee who notices an imbalance can exploit it without reading anything. Asking a fixed solver each item and recording which position it chose:

bank	A	B	C	D	χ^2 vs uniform (3 df)
RAC	**0.900**	0.050	0.050	0.000	**90.4**
naive	0.225	0.550	0.225	0.000	24.6
RAC, after balancing	0.200	0.175	0.300	0.325	**2.6**

Ninety percent of the RAC bank's keys sit in position A, against a critical value of 7.81 at $p = .05$. An examinee who answers A to everything scores 90% without reading a stem, and neither bank ever places a key in D. This is a property of the solver only if the solver is not reading the items, so we separated the two by permutation: ask each item twice, once as written and once with its options reordered, and record which option *text* is chosen. The answer follows the content in 95% of RAC items and 100% of naive items, and stays on the same letter in only 17.5% and 30%. The solver reads; the imbalance is in the bank.

Why the method cannot see this. Key position is not a semantic property, so it is invisible to every measure in this paper — embedding diversity, *n*-gram diversity, Vendi, coverage and the enemy-item radius are all exactly as favourable on a bank whose key is always A as on a balanced one. It is invisible to the axis set for the same reason: all seven axes condition on item content, and even *distractor logic* governs what the distractors are like rather than where the key sits among them. And it is structural rather than accidental — a generator asked for a stem and four options writes the answer it has in mind first and builds distractors around it, so A is where the key lands by default.

The repair needs no answer key. Permuting each item's options uniformly at random moves the key with its own text from whatever position it held, so the bank becomes uniform by construction; options whose text refers to a position (”none of the above”, ”both A and B”) are detected and left as written, which affects 37 of 1,999 items. Applied to the RAC bank this takes χ^2 from 90.4 to 2.6 — indistinguishable from uniform — while content-tracking is unchanged at 95%.

The general form is the sharpest limitation in this paper. **A diversity objective defined over an embedding is blind to any property of the artifact the embedding does not encode, including properties that decide whether the artifact can be used at all.** Difficulty is the same story in the same domain: it is the parameter an item bank exists to span, it is latent rather than semantic, and nothing in the objective can see it. Measuring diversity well is not the same as measuring fitness for purpose, and the gap between them is not visible from inside the objective.

5. Steering through the proxy

5.1 Images: closing the loop where the product lives

The vision-steered arm closes the loop where the product lives. It renders a bounded sample of accepted instructions, embeds them with CLIP, and feeds two things back into the text-side loop: a least-squares map from the crowded *image* directions into instruction-embedding space, so the text-side orthogonality term can push away from visual redundancy it cannot itself perceive; and mined *visual* attractors from the vision judge, appended to the same ledger as the textual ones. It is the paper's mechanism applied one level down — the ledger already repels against what the model keeps saying, and now also against what it keeps showing — and it is the best arm in the companion paper's competitive comparison (its the companion paper's §7.2).

The margin is measurable in the space the loop optimizes and visible on the page. At matched n = 200 in the DALL·E domain, the text-only RAC arm scores centered Vendi 77.67 and median nearest-neighbour distance 0.171; the vision-steered arm scores 91.97 and 0.192, an 18% gain in centered Vendi, and leads every column of the seven-arm comparison — distinct-2 0.699 against 0.660, 4-gram self-repetition 0.040 against 0.068, n-gram Vendi 170.9 against 167.0. The contact sheet of §4.2 shows the same thing without a number: rows 3–4, steered on both the text and the vision side, move medium, palette, register and composition where rows 1–2 return one visual mode.

5.2 Audio: cross-modal steering through an audio embedder

At matched n, matched prompt length and the same generator, the conditioned music arm reaches centered Vendi 43.03 against naive's 22.99 (1.87×), halves 4-gram self-repetition (0.140 against 0.291), raises distinct-2 (0.545 against 0.348) and holds roughly three times the room between nearest neighbours (0.079 against 0.027), on 100 prompts per arm with no exact duplicates in either — so the effect is semantic. On 100 rendered Lyria instrumentals generated by cross-modal steering with zero rejection, 100% of tracks verify as instrumental in CLAP space and mean-centered CLAP Vendi is 17.43 against 11.5 for naive prompts and 14.1 for axis-conditioned prompts under the same embedder.

The rendered arm was steered through CLAP and is measured in CLAP. §4.3's alignment table says the same loop through MuQ-MuLan would have a steering gradient nearly four times as strong on the naive arm (0.68 against 0.18), which is the experiment we would run next; the prescription — steer music with MuQ-MuLan rather than CLAP — is §4.3's, and the loop is the image one of §5.1 with the audio tower in place of CLIP.

*5.3 Best-of-*K *at the render: select on the channel the objective cannot see*

The companion paper measures best-of-K on the written candidate (its §8.4): selecting on the gap alone raises the minimum nearest-neighbour distance by 20% in three of three seeds, at a cost of a third of a craft point. The same question at the *render* has a sharper answer, because the two channels have very different effective dimensions. Best-of-K on the written candidate happens before the handoff and §6 locates the loss after it, so we also ran K = 3 renders behind each of 60 fixed instructions and kept the render maximizing min distance to the accepted set — once in CLIP, once in optical statistics, over the same candidates. Selecting on optics raises pixel min NN from 1.039 to 1.292 (1.24×) and the fifth-percentile NN by 0.336 (paired subsample, 95% CI [+0.128, +0.488]), **and costs nothing in the channel the method optimizes**: CLIP min NN moves by −0.001 with the interval [−0.029, +0.033] tight around zero. Selecting on CLIP raises CLIP min NN by 0.040 ([+0.025, +0.071]) and moves optics not at all detectably. Both gains lie on the $K^{1/m}$ curve — 1.24× at $K = 3$ implies $m \approx 4.4$, and 1.04× implies $m \approx 17$ — so render-side selection is oversampling rather than a new mechanism. What the dimensions add is an allocation rule: oversampling buys far more in a low-dimensional channel than in a high-dimensional one, and **if a pipeline is going to pay for extra renders, it should select them on the channel the objective cannot see.**

6. Conditioning across the seam

The companion paper audits whether a commanded axis level shows up in the artifact (its §8.1), by manipulation rather than correlation: hold a base spec fixed, set one axis to each of its levels in turn, generate, and read the displacement, centring within base so that only variation produced by *setting* the axis contributes, with a null that shuffles levels within each base. Two instruments read each artifact — *identifiability*, whether CLIP(ℓ) is the nearest of the axis's level descriptions to an image generated under level ℓ, against a chance rate of 1/|levels|; and *separation*, the η^2 of artifact embeddings grouped by commanded level under a permutation null — and a third channel that lives outside the embedding the method optimizes: pixel statistics for images, prosody for poems. On poems every axis is realized on at least one channel. On images three of seven axes identify their level at or below chance, two are undetectable on both channels, and the axis the attribution scoring ranked *first* is the least realized in the corpus. The image domain is where the proxy question becomes a conditioning question, because the image pipeline has a seam the poem pipeline does not.

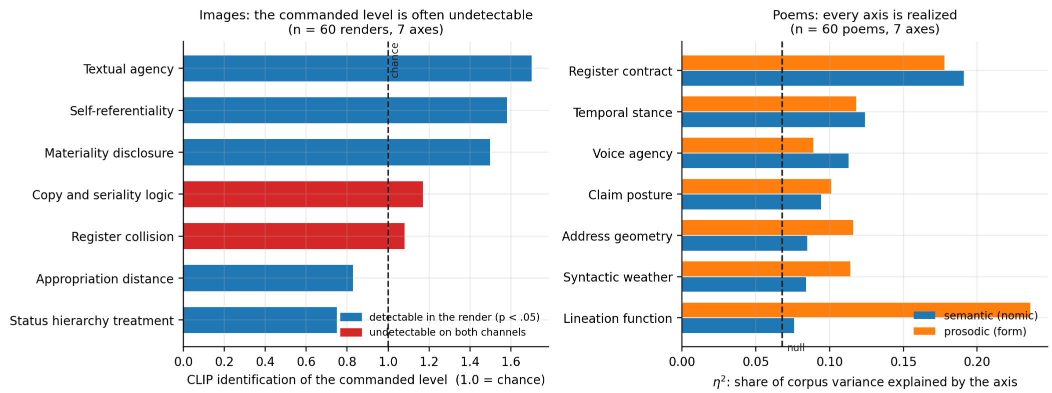

Figure 3. Axis realization measured on the artifact. Left: on 60 max-min renders, three of seven axes identify their commanded level at or below chance, and two are undetectable on both the CLIP and pixel channels. Right: on 60 poems, every axis is realized — on the semantic channel, the prosodic channel, or both. Dashed lines mark chance and the permutation null.

Images, by intervention: the loss is at the render, not in the writing. The image pipeline has two stages — the model that proposed the axes writes an instruction, and `gpt-image-1-mini` renders it — so the same manipulation can be read on both sides of the handoff. Over all eleven axes at three bases and five levels, embedding the written instruction with CLIP, the render with CLIP, and the render's optics separately: nine of eleven axes are realized in the **written instruction** at $p < 0.05$ (mean $\rho = 0.148$), against four of eleven in the **render** (mean $\rho = 0.096$). Ten of the eleven attenuate, a mean loss of 35% of the text-side effect (Wilcoxon signed-rank $p = .005$; paired t, $p = .003$, both two-sided). **Conditioning reaches the instruction and loses a third of its grip at the render.** The image domain's weak realization is therefore not a failure of the axis calculus to produce usable conditioning — the conditioning is there, in the prompt, and measurable — but of that conditioning to survive a generator that never saw the axis set.

Part of the apparent loss is the instrument rather than the renderer, and this is the §4.1 lesson arriving from the other direction. The four perceptual axes attenuate most in CLIP and are precisely the axes CLIP is least equipped to register: measured on pixels instead, *Lighting direction* reads 0.229 against 0.092 in CLIP, *Shot scale* 0.252 against 0.105, and *Motion and temporal blur* 0.203 against 0.069. An axis commanding optics moves the optics and barely moves a semantic embedding.

The seam explains the split. The calculus is identical across the two domains, the axes are comparably abstract, and the budgets match. What differs is where the conditioning has to travel. A poem is written by the same model that proposed the axes, so the conditioning never leaves the system that understands it. An image axis is written by that model into a prompt and handed to a different model, which never saw the axis set, shares no latent space with it, and is under no obligation to honour a distinction like *copies outrank an absent original*. the companion paper's §4.2 measured this seam from the outside as a proxy gap; the intervention measures it from the inside. **Conditioning strength degrades at the boundary where the generator changes hands**, so the modality-agnostic claim holds for the calculus — which needs only an embedding — and not for the conditioning, which needs a generator that reads the same language the axes are written in.

7. Scoring axes in the artifact's space

All four factors are computed from *level vectors*: one embedding per level of each axis. In the text-only setting those are embeddings of the level descriptions — what a level *claims* it will do. Once the artifact can be embedded, the natural improvement is to replace them with the centroid of everything actually produced under each level: what the level *did*. That substitution is what makes the scoring empirical, and on its own it is wrong.

A centroid computed that way is a **marginal mean where the score needs a partial effect**. Specs are sampled independently, so every other axis varies inside each level group; with sixty items, seven axes and five levels each, a group holds about twelve items whose spread is mostly produced by the other six axes. The estimator cannot tell "this axis moves the artifact" from "this axis co-occurred with movement", and nothing in the loop ever intervenes on one axis while holding the rest fixed.

The consequence is not a modest loss of precision. It is a **systematic inversion**, and the mechanism is our own transversality term. An axis that genuinely moves the artifact fills the corpus with variance along its own direction; the occupied eigenspace absorbs that direction; and transversality — the fraction of an axis's variation lying *outside* the occupied span — collapses toward zero for precisely that axis. An axis the generator ignores contributes only isotropic noise, little of which the occupied basis captures, so its transversality stays high. Because promise is multiplicative, the effective axis sinks and the inert one rises. **The better an axis is, the harder the score punishes it.** The repair — partial effects from one additive model over all axes, a null-corrected realization factor ρ, and the interventional probe when a generation budget can be spent on measurement — is the companion paper's §5.3, and it is what makes artifact-space scoring usable: with it the strong axis ranks first on all 24 seeds of a ground-truth test where the attribution form ranked the inert axis first in 17 of 24. What belongs here is the consequence for a proxy. Level vectors read in the artifact's own embedding are only as good as that embedding's grip on the axis. §6 shows the four perceptual image axes attenuate most in CLIP and are precisely the axes CLIP is least equipped to register — *Lighting direction* reads 0.229 on pixels against 0.092 in CLIP — so an axis commanding optics can be scored as inert in the very space that was adopted to make the scoring honest. Where a designed feature channel exists for the artifact, the realization factor should be computed there as well, and the axis kept if either channel finds it.

8. Practice

The prescriptions are short, and each is attached to the measurement that motivates it.

1. **Name the embedder, and publish the pairwise-similarity distribution it operates on.** Per-arm audio verdicts flip between embedders and each nominates a different most-redundant pair (§4.3); mean pairwise cosine was 0.883 in one text corpus and 0.444 in another under the same embedder. A diversity number without its embedder and its similarity distribution is not a number.
2. **Count exact duplicates before computing anything, and report the literal and the latent separately.** They move in opposite directions in the same corpus (§4.5), and a 74% duplicate rate makes every distance statistic a statistic about duplication.
3. **Measure at the level of the artifact you ship.** Text-embedding similarity explains 14.5% of the variance in whether rendered images look alike, pooled across seven arms, and 8.2% within-arm (§4.2). If the artifact is rendered, the objective belongs in the space the artifact occupies.
4. **Audit in at least one channel the objective never optimizes**, with a distance statistic rather than a spectral one (§4.1). It is the only measurement that cannot be a restatement of the selection rule, and it is where the strongest positive result and the largest undetected defect in this line of work were both found.
5. **When a render is in the loop, steer through it.** Render a bounded sample, embed it in the artifact's space, and bridge its crowded directions back into the text space the loop optimizes (§5.1); steer audio through an embedder whose text and audio towers agree (§5.2).
6. **Spend extra renders on the low-dimensional channel.** Best-of-K buys $K^{1/m}$; selecting renders on nine pixel statistics buys 1.24× at $K = 3$ where selecting on CLIP buys 1.04× (§5.3).
7. **Measure axis realization at the artifact, by intervention, wherever the generator changes hands** (§6). Where the seam exists, a third of the conditioning does not cross it.
8. **Balance by construction what the embedding cannot see.** Key position, difficulty, compositional template: none is encoded, none is optimized, and each decides usability. A uniform permutation of options needs no answer key and takes χ^2 from 90.4 to 2.6 (§4.7); a structural ban block halves palette twins (§4.6).
9. **Read the output.** A corpus can be diverse along every dimension you thought to measure and uniform along the one you did not.

9. Limitations

- **One text embedder.** All text geometry is `nomic-embed-text` with cosine distance. §4.4 shows the *ranking* of methods survives four representations; whether the *magnitudes* transfer is untested, and the companion paper's coverage and packing numbers inherit that.
- **A joint embedder varies both towers at once.** §4.3 changes one knob — the model supplying both the text tower and the audio tower — so it shows that the alignment moves and cannot say which side moves it. Crossing text-side and artifact-side encoders independently on fixed bytes is the decomposition this paper does not contain.
- **The renders carry no logged seed.** The image endpoint exposes no seed parameter, so no render in this paper can be regenerated, and render stochasticity cannot be separated from the quantity being measured. Every artifact-space number here — the proxy correlations of §4.2, the realization of §6, the render selection of §5.3 — is a statement about an unlogged seed distribution, and a repeat-draw floor at a logged seed is the missing measurement.
- **Several image results are single-seed.** The image arms are one seed each at n = 60; the vision-steered arm's margin in §5.1 is one run at n = 200; the audio steering arm is one run of 100 tracks.
- **Judge bias is unmeasured.** Vision and language-model judges name attractors, classify blueprint areas and read device forms; none is calibrated against human raters, and a judge that shares the generator's blind spots would miss exactly what the generator misses.
- **The structural signatures are deliberately dumb.** A 16×16 luminance map, a tiling score and a hue histogram find tilings and palette twins and nothing subtler; they are a floor on what the embedding misses, not a measure of it.
- **Steering closes part of the gap and not all of it.** Literal-space bans halve palette twins and improve the coverage objective by 21% and do not close the structural deficit; the vision-steered arm is the best arm in the image domain and still renders through a model that never saw the axes.

10. Conclusion

An embedding oracle is the only thing the loop can see, and it is always a proxy. In the two domains where the artifact is rendered by a model the embedder never meets, the proxy explains a seventh of the variance in what the reader receives, whether a prompt can be steered at all depends on which embedder is asked, and a third of the conditioning does not survive the handoff. In the domains where the artifact is text, the embedding is still blind to metre, to compositional template, and to where an answer key sits — properties that decide whether the corpus can be used.

None of this is an argument against the method; it is an argument about where the method's numbers are true. The extension that follows is to treat the proxy as a proxy: render a sample and steer through the artifact's own space, spend oversampling budget on the channel the objective cannot see, audit realization by intervention at the seam, and balance by construction what no embedding encodes. Under those rules the ranking the companion paper reports is robust — first under four representations, three of which the method never optimized, at a mean Kendall τ of +0.83 — and the gap between a diverse corpus and a usable one is at least visible, which is the precondition for closing it.